5G Security Identifiers

SUCI & NAI-Based SUPI – A Technical Reference

Part 1: SUCI – Subscription Concealed Identifier

What is SUCI?

SUCI (pronounced *su-see*) is a **privacy-preserving identifier** introduced in 5G NR (New Radio) to protect the subscriber's permanent identity (SUPI) over the air interface. It was introduced to fix a major privacy vulnerability in 4G/LTE, where the IMSI could be sent in plaintext and intercepted by IMSI catchers (e.g., Stingrays).

Background: The Problem with 4G

In LTE, the **IMSI** (International Mobile Subscriber Identity) was transmitted in cleartext during initial attach procedures, making it easy to track subscribers using rogue base stations. 5G addresses this with SUCI.

SUPI vs. SUCI

Term	Full Form	Description
SUPI	Subscription Permanent Identifier	The permanent identity (equivalent to IMSI). Never sent in plaintext over t
SUCI	Subscription Concealed Identifier	A one-time, encrypted version of the SUPI sent over the air.

Structure of SUCI

A SUCI is constructed from the following components:

```
SUCI = SUPI Type | Home Network Identifier | Routing Indicator | Protection Scheme ID |
Home Network Public Key ID | Scheme Output (encrypted MSIN)
```

- **SUPI Type**: Indicates IMSI-based or NAI-based SUPI
- **MCC + MNC**: Mobile Country Code + Mobile Network Code (sent in plaintext for routing)
- **Routing Indicator**: Helps route to the correct AUSF/UDM (sent in plaintext)
- **Home Network Public Key ID**: Identifies which public key was used
- **Protection Scheme ID**: The ECIES profile used
- **Scheme Output**: The encrypted MSIN (the sensitive part)

Only the **MSIN** (Mobile Subscription Identification Number) is encrypted. MCC/MNC and Routing Indicator remain visible for routing purposes.

Encryption: ECIES Profiles

SUCI uses **Elliptic Curve Integrated Encryption Scheme (ECIES)**:

Profile	Curve	KDF	MAC/AEAD

Profile A	Curve25519	HKDF-SHA-256	AES-128-CTR + HMAC-SHA-256
Profile B	NIST P-256	HKDF-SHA-256	AES-128-CTR + HMAC-SHA-256
Null-scheme	—	—	No encryption (specific cases only)

How SUCI is Generated

1. The **USIM** (SIM card) or the **ME** (Mobile Equipment) generates the SUCI. Per 3GPP TS 33.501, SUCI generation *should* happen on the USIM for better security.
2. A fresh **ephemeral key pair** is generated each time, ensuring SUCI is different every registration (unlinkability).
3. The MSIN is encrypted using the **home network's public key** (stored on the USIM).

SUCI Usage Flow

```
UE --[Registration Req (SUCI)]--> gNB/AMF --[Authentication Req (SUCI)]--> UDM/SIDF
[SIDF decrypts SUCI → SUPI] UE <--[5G-GUTI assigned]--------- gNB/AMF <--[Auth Response
(SUPI-based)]-- UDM/SIDF
```

After the first authentication, the network assigns a **5G-GUTI** (Globally Unique Temporary Identifier) for subsequent communications, so SUCI is only needed on initial registration or when no valid GUTI exists.

SIDF – The Decryption Entity

The **Subscription Identifier De-concealing Function (SIDF)** resides within the **UDM** (Unified Data Management) in the home network. It is the only entity with the private key to decrypt SUCI back to SUPI.

Key Security Properties

- **Identity Privacy**: SUPI never exposed over the air
- **Unlinkability**: Each SUCI is unique (ephemeral keys), preventing tracking
- **Home Network Control**: Only the home operator can de-conceal the identity
- **Replay Protection**: Stale SUCI cannot be reused to identify a subscriber

Relevant 3GPP Specifications (SUCI)

- TS 33.501 – Security architecture and procedures for 5G
- TS 23.003 – Numbering, addressing, and identification
- TS 24.501 – NAS protocol for 5GS

Part 2: NAI-Based SUPI

What is NAI?

NAI stands for **Network Access Identifier**, defined in **RFC 7542** (IETF). It is a standard format used to identify users in network access scenarios, commonly used in EAP (Extensible Authentication Protocol) frameworks. The general format is:

```
username@realm
```

For example: `device-001@5gc.mnc001.mcc310.3gppnetwork.org`

Two Types of SUPI

Type	Format	Usage
IMSI-based SUPI	15-digit number (MCC + MNC + MSIN)	Traditional SIM-based subscribers
NAI-based SUPI	username@realm (RFC 7542)	Non-SIM devices, private networks, enterprise, IoT

The **SUPI Type** field in the SUCI indicates which form is being used: value **0** for IMSI, value **1** for NAI.

When is NAI-Based SUPI Used?

- **Non-3GPP Access (Wi-Fi / N3IWF)**: When a UE connects to 5G Core via untrusted non-3GPP access (e.g., Wi-Fi through N3IWF), NAI-based SUPI is commonly used.
- **Private 5G / Enterprise Networks**: Enterprises running private 5G networks may use NAI-based identifiers aligned with their existing identity infrastructure.
- **Soft SIM / iSIM / eSIM with EAP-based Authentication**: Devices using EAP-AKA' or EAP-TLS authentication naturally align with NAI formatting.
- **IoT / Machine-Type Communication (MTC)**: IoT devices without traditional SIM credentials can be identified using NAI-based SUPI.
- **5G LAN / Network Slicing**: Specific slices or verticals may provision subscribers using NAI identities.

Format of NAI-Based SUPI

Per 3GPP TS 23.003, the NAI-based SUPI takes the form `username@realm`, where:

- **username** – A subscriber-specific identifier (operator-defined)
- **realm** – Typically the home network domain name, e.g., `5gc.mnc001.mcc234.3gppnetwork.org`

Example:

```
device-001@5gc.mnc001.mcc310.3gppnetwork.org
```

SUCI Construction for NAI-Based SUPI

When the SUPI is NAI-based, the SUCI construction differs from IMSI-based:

```
SUCI = SUPI Type (1) | Scheme Output | @ | realm
```

- **SUPI Type = 1** signals NAI-based identity
- The **username part** of the NAI is encrypted using ECIES (sensitive portion)
- The **realm part** (`@realm`) remains in plaintext for routing — just like MCC/MNC in IMSI case

NAI-Based SUPI in the Authentication Flow

```
UE (NAI SUPI) --[Reg Req (SUCI, SUPI Type=1)]--> AMF AMF --[Nudm_UECM
(SUCI)]-------------> UDM/AUSF [SIDF decrypts SUCI --> NAI-based SUPI] [AUSF runs
EAP-AKA' or 5G-AKA] UE <--[Auth Challenge]---------------- AMF UE --[Auth
Response]----------------> AMF UE <--[5G-GUTI assigned]-------------- AMF
```

The authentication method used with NAI-based SUPI is typically **EAP-AKA'** rather than 5G-AKA, though 5G-AKA can also be used depending on operator configuration.

Key Differences: IMSI-Based vs. NAI-Based SUPI

Aspect	IMSI-Based	NAI-Based
Format	Numeric (MCC+MNC+MSIN)	username@realm
SUPI Type value	0	1
Primary use case	Traditional SIM subscribers	Non-SIM, enterprise, IoT, non-3GPP access
Encrypted part in SUCI	MSIN	Username portion
Plaintext routing info	MCC + MNC + Routing Indicator	Realm
Auth method	5G-AKA or EAP-AKA'	Typically EAP-AKA' or EAP-TLS
Credential storage	UICC/USIM	Flexible (SE, TEE, software)

Relevant Specifications (NAI-Based SUPI)

- 3GPP TS 23.003 – NAI-based SUPI definition and format
- 3GPP TS 33.501 – SUCI construction for NAI-based SUPI
- RFC 7542 – NAI format (IETF)
- 3GPP TS 33.402 – Security for non-3GPP access (EAP methods)

Reference: 3GPP TS 23.003 | TS 33.501 | TS 24.501 | RFC 7542 | TS 33.402